

Sandeep Kumar Chittimalli

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Research Interest

Artificial Intelligence, Machine Learning, Data Science, Remote Sensing, Ocean Science, Water Quality, Earth Observation Sensors Cal/Val, Image Processing, Satellite Image Processing

Work and Research experience

Synectics for Management Decisions Inc / Contractor@NSF **May 2021 - Present**
Data Scientist

- Built and administrate Data Science Desktop Environments to assist NSF Data Science User Community in running crucial Machine Learning (ML) models that play a critical role in decision making, as well as enabling the applications of Artificial Intelligence at the foundation.
- Conducting research and developing Proposal Topic search tool to identify keywords and related topics from a cluster of Remote sensing proposal data particularly in the fields of Ocean Science, Land-cover, Vegetation using word-embedding algorithms such as GloVe, Word2Vec, Sagemaker BlazingText and searchmetrics.
- Building various AI/ML tools to support Ocean Science Scientist group at NSF.
- Enhancing current natural language processing (NLP) methods such as word vector, similarity scores, and proposal binning to create new applications using available Earth-observation proposal data within NSF.
- Developed a cloud-native Data Science Environment and infrastructure to assist analytics and the transition of current machine learning projects and data science applications to a cloud-based data processing environment.
- Developing machine learning, predictive, forecasting models for various NSF proposal data. Assisting in thought leadership and training staff on analytics, data science, machine learning, and other related topics.
- Using Machine and deep learning techniques to improve accuracy of current Optical character recognition (OCR) algorithm which extracts text from images.
- Research, document end to end, prepare proof-of-concept reports (POC), and employ new data science research tools to the data science user's community that helps to derive unique findings; I ensure all these research reports are created using credible qualitative, and quantitative methodologies based on key insights from historical and current studies.
- Utilizing Cloud-based services, Elastic Computing Cloud (EC2), Elastic MapReduce (EMR), AWS Comprehend, Sagemaker, Elastic Kubernetes Service (EKS), Redshift to build and deploy seamless Machine Learning business products.
- Assisted in developing advance analytical tools such as Open Data Applications, Data Dictionaries, Data science engineering pipelines, and built Data Visualizations apps using R-Shiny, Jupyter Hub Voila server, Streamlit, Plotly Dash, Bokeh, Tableau, and Oracle Business Intelligence Enterprise Edition (OBIEE), etc.
- Automated and Optimized routine scripts such as Proposal download report process which take longer time when users submit bulk requests.

Capital One Senior Data Analyst

Feb 2021 - May 2021

- Created Natural Language Processing (NLP) modeling tools for an Artificial Intelligence (AI) driven chatbot in Python.
- Designed various computing platforms, multiprocessing data analytic pipeline tools in python on AWS Cloud that consumed massive structured and unstructured data from various sources and stored in Snowflake. Finally applied machine learning algorithms to solve various business problems.
- Part of the California consumer privacy Act (CCPA) project, I developed automated scripts to mask any personal information (PI) and non-personal information (NPI) in the data. Redacted and labeled PI items in Data Lakes using a machine learning-based natural language processing (NLP) tool like Amazon Comprehend and used entity scores data for each detected element for additional validation.

Synectics for Management Decisions Inc / Contractor@NSF Data Scientist

Oct 2020 - Feb 2021

- Evaluated various data visualization tools such as Voila, Bokeh, Plotly Dash Streamlit to deploy AI/ML apps in python for collaborative sharing and prepared a proof of concept (POC).
- Built and managed on-premises Data Science Desktop environments that include installation, configuration of the tools that support to latest advance analytics scripts like Python, R, and Maintain a centralized process to run Data science analysis.
- Provided initial document for smooth migration of On-premises Data Science environment to cloud, support and serve to give best solutions to perform data science tasks on various cloud platforms and to reduce impact for data science users.
- Automated scripts to get R and python data science users activity and kill data science pipelines those running for hours and consuming a lot of system resources and implemented an automated script which gives the usage of Data Science Desktops and alerts when it exceeds minimum thresholds.

Data Analyst Capital One

July 2018 - Oct 2020

- Developed a daily alert to detect anomalies in digital messaging data volume utilizing Machine learning algorithms that helped many product owners to save time and capture failure events.
- Creating images-based decision plots and generated daily Data quality monitoring reports using Capital One AI Chatbot data.
- Implemented A/B testing to test the impact of promotional offers, messaging, and email campaigns.
- Created various machine learning models for business needs and deployed them on AWS cloud. Optimized various data science pipelines and helped team transition between various systems.
- Developed various Time series and Predictive modeling for AI chatbot data. Actively participated in developing NLP tools for modeling Text analytics.
- Built a library of models and various packages in python and implemented reusable scripts and introduced new skills to the team to produce consistent-streamlined business intelligence results.
- Built automated daily KPI reporting workflow and pipelines that gather and aggregate data from many sources. Lead and implemented many data-pipelines using AWS lambda and

Python, which saved business in terms of time and cost.

- Assisted team members in Running big data analytic jobs, data science applications in various computation engines such as Databricks, spark, Elastic MapReduce (EMR) and Elastic Cloud (EC2).
- Built data pipelines, data quality, and visualizations for a variety of sources such as Kafka topics, Streaming Data Platform (SDP), S3, analytical raw data, and so on.
- Developed performance-optimized Tableau dashboards to monitor product performance, allowing business stakeholders to self-serve data needs, and improving analytics productivity.

USRA / Contractor @NASA GSFC Research Analyst and Visiting Scientist

Feb 2017 - July 2018

- Developed and performed Intercomparison of satellites (Landsat-Sentinel) and Cross-Cal terms of Radiance and reflectance for aquatic applications.
- Learned and used MODTRAN for Atmosphere corrections and Relative Spectral Responses (RSR) corrections.
- Developed Python and Linux scripts to use NASA's Sea-viewing Wide Field-of-view Sensor (SeaWiFS) Data Analysis System (SeaDAS) tool and generate level-2 products.
- Developed atmospheric corrections using SeaDAS for Landsat series, Sentinel series, MODIS, VIIRS satellite data.
- Developed a Landsat/Sentinel acquisition planner in Python Django, utilizing Two-line Element (TLE). This tool helped field associates to forecast satellite overpasses and prepare in advance in order to collect and compare ground (field) water measurements with satellite data.
- Implemented algorithms and performed Cross-calibration and Inter-comparisons of various satellite sensors data to use for Land and aquatic applications.
- Developed various Image process techniques in Python for Landsat, Sentinel satellites level-1 and level-2 products and analyzed various satellite preliminary datasets on a pixel level using image processing techniques.
- Studied and applied various Machine Learning concepts to detect outliers in satellite spatial and temporal data.
- Developed an algorithm and automated scripts that leveraged Landsat/Sentinel data to detect algal blooms in aquatic data. Using SeaDAS (Level 2) historical data, it captured dynamic fluctuations within Chl-a and TSS. This concept prompted water managers to make swift choices regarding where satellites detected anomalies and suspicious blooms.
- Utilized MAD Z-score, ARIMA, SVM, Grubbs and a few machine learning techniques to detect anomalies in spatial and temporal datasets.
- Developed automated scripts and Linux batch scripting for bulk satellite data processing and batch processing of various levels of satellite data.
- Implemented predictive models to predict future anomaly events based upon spatial and temporal time series satellite data.
- Maintained scripts and ran SLURM jobs on High Performance computing (HPC) machines and helped the team on various projects and purposes.
- Utilized latest scripting languages such as Python, R, Statistics, and machine learning algorithms and studied global water lakes satellite data to study retrieval improvements.
- Maintained a good relationship with various partners and Stakeholders around the globe.
- Automated scripts and developed tools to download data from Amazon S3 buckets, United States Geological Survey (USGS) archive which saved project time.
- Paper publications, Presentations, and talks.

Acknowledgements, Contributions and Projects Supported

- Sentinel-2 Multispectral Instrument (MSI) data processing for aquatic science applications: Demonstrations and validations. Remote Sensing of Environment (2017)
Link : [Acknowledged in page:56](#)
- Assessment of atmospheric correction algorithms for the Sentinel-2A Multispectral Imager over coastal and inland waters. Remote Sensing of Environment (2019)
Link: [Acknowledged in page:282](#)
- Predicting microcystin concentration action-level exceedances resulting from cyanobacterial blooms in selected lake sites in Ohio. Remote Sensing of Environment (2020)
Link: [Acknowledged in page:7](#)
- Aquatic Science and Applications Enabled by the Landsat 8 Operational Land Imager.
Link: <https://agu.confex.com/agu/os18/meetingapp.cgi/Paper/303949>
- Geospatial Monitoring of the SDGs Sharing experiences on indicator 6.6.1 on freshwater related ecosystems, and exploring opportunities for better monitoring of the land-water-ocean nexus
Link: https://ggim.un.org/meetings/2017-Mexico/documents/NASA_EO4SDGs.pdf

South Dakota State University Graduate Research Assistant

May 2015 – Dec 2016

- Implemented Radiometric algorithms (Radiance and Reflection based calibrations) for whole Landsat satellites data archive (40 years of Landsat data) to Support USGS.
- Used EO-1 Hyperion (hyperspectral imager) to build spectral band adjustment factor (SBAF) for other Landsat/Sentinel datasets.
- Implemented automated scripts in Matlab/Python to extensively use Pseudo invariant calibration sites (PICS) for cross calibrations and validations of satellite datasets.
- Worked on Landsat MSS series and gained exceptional knowledge in this area.
- Improved uncertainty of the past 4 decades and future upcoming data products of Landsats. Implemented image processing algorithms using MATLAB and Python.
- To find more calibration Region of Interest (ROI) sites analyzed Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI) and Bidirectional reflectance distribution function (BRDF) characteristics.
- Developed multiple statistical tests in Python and MATLAB to validate results.
- Developed a Python application to find simultaneous Landsat satellite overpasses data which made it easy for calibration and save time.
- Documented the problems and progress of project most accurately.
- Developed a number of automated scripts to download Landsat data from United States Geological Survey (USGS) archive.
- Presented research and calibration results in Landsat Science conference meetings. (<https://landsat.usgs.gov/lst-meetings-presentations-search>)
- Paper publications, Presentations, and talks.

South Dakota State University Graduate Teaching Assistant

Jan 2015 – May 2015

- Handled lab sessions on Digital Systems.
- Taught Digital Systems lab projects using PIC Micro Controller, Register-transfer level (RTL) designs on Field Programmable Gate Array (FPGA).
- Taught students Verilog coding and algorithms to develop projects.
- Performed routine jobs such as proctoring exams and grade exam papers and clarifying undergrad subject questions.

Young Minds Technology Solutions

Aug 2012 – Aug 2014

VLSI and Imaging Engineer

- Designed image processing projects according to requirements using Verilog and Matlab.
- Implemented face detection algorithms using Verilog and Matlab and deployed them on Field Programmable Gate Array (FPGAs).
- Gained good understanding in Machine learning concepts and computer vision algorithms.
- Implemented and worked with IRIS data, K-Means Clustering, Support vector machine. (SVM), and Self-organized map (SOM's) in Python and Matlab.
- Implemented image processing algorithms using Gabor filters for disease detection, motion detection algorithms, Gaussian filters, Haar classifiers, and Adaboost algorithms in Matlab and Verilog.

Teaching, Training and Academic Experience

Jan 2013 – May 2013

(JB Institute computer Technology, Chadalawada Ramanamma Engineering College)

Guest Lecturer (Part of Young Minds Technology Solutions field work)

- I taught courses such as Digital Image Processing, Switching Theory and Logic Design (STLD), Verilog, Very High-Speed Integrated Circuit (VHSIC) Hardware Description Language (VHDL), Matlab.
- Helped graduate students to implement projects using tools such as FPGA, VHDL, Verilog, MATLAB, and Image processing.

Intern and Project Trainee

Apollo Computing Laboratories (P) Ltd Jan 2012 – Mar 2012

Intern Trainee

- Part of the Final year Project and thesis work developed a satellite communication data sync prototype called 'Frame Synchronizer' using Verilog language.

Academic Credentials

Master's in Electrical Engineering (Emphasis in Image Process & Remote Sensing)

South Dakota State University, Brookings, South Dakota, USA

Jan 2015 – Dec 2016

Cumulative GPA (3.3/4), Core courses and Seminar cumulative GPA (3.6)

Project 1: 3D Object and Image Modelling by Phase Shifting Profilometry and Gamma Correction.

Project 2: Canon EOS 450D Radiometric Calibration.

Master's Thesis: "Reflectance-Based Calibration, and Validation of Landsat Satellite Archive" (Available at :

<https://openprairie.sdstate.edu/cgi/viewcontent.cgi?article=2104&context=etd>)

Master of Technology in VLSI System Design

Jawaharlal Nehru Technological University – Hyderabad (India)

Mar 2013 – Nov 2014

Cumulative GPA (3.90/4)

Bachelor of Technology in Electronics and Communication Engineering

Jawaharlal Nehru Technological University – Hyderabad (India)

Sep 2008 – May 2012

Cumulative GPA (3.70/4)

Project: Image Enhancement and Noise Removal using Adaptive Bilateral Filters.

Thesis: Frame Synchronizer (A module which synchronizes and captures data coming from satellites).

Journals, Conference Papers & Technical reports

Journals:

1. Nima Pahlevan, Sandeep K. Chittimalli, Sundarabalan V. Balasubramanian, Vincenzo Vellucci, "Sentinel-2/Landsat-8 product consistency and implications for monitoring aquatic systems," *Remote Sensing of Environment*, Volume 220, 2019, Pages 19-29, <https://doi.org/10.1016/j.rse.2018.10.027>
2. Sandeep K. Chittimalli, "Cross-Lake Transferability of Sentinel-2 Turbidity Retrieval Using Leave-One-Lake-Out Validation," *IEEE Geoscience and Remote Sensing Letters*, 2026. (Under Review)

Conference papers:

1. Cibelex Teixeira Pinto, Sandeep K. Chittimalli, Larry Leigh, Timothy Ruggles, Dennis L. Helder, "A reflectance-based cross calibration of the Landsat sensors," *Proc. SPIE 10402, Earth Observing Systems XXII*, 104021C, 5 September 2017
2. Sandeep K. Chittimalli et al., "OpenCalValPlan: A Reproducible System for Multi-Sensor Calibration, Validation, and SNO Planning," *IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2026)*, Washington, DC, 2026. (Accepted)

Technical Reports, Papers, Books and Thesis

1. SDG Indicator 6.6.1. Change in the extent of water related ecosystems over time. Methodology for the use of satellite-based Earth observations datasets and respective tools for country level reporting, October 2017, NASA, University of Maryland, UN Environment, Earth Observation Sustainable Goals, Group on Earth Observations.
2. Sandeep K. Chittimalli, "Reflectance-Based Calibration and Validation of the Landsat Satellite Archive," South Dakota State University, 2016.
3. Earth Science Information Partners & American Geophysical Union. (2025). Data Help Desk @ AGU 2025, New Orleans, Louisiana, USA. Zenodo. <https://doi.org/10.5281/zenodo.18089255>
4. Sandeep K. Chittimalli, "Secure and Decentralized Architectures for Industrial IoT and Digital Twin Ecosystems," CRC Press, Taylor & Francis, 2026. (Abstract Accepted, Full Paper Under Review)

Industry Publications & Technical Articles

1. PII Scanning Tools for Big Data, Sandeep Kumar Chittimalli, Publication date: 02/2024
Link: <https://smdi.com/navigating-the-pii-minefield-tools-and-tactics-for-ultimate-data-security/>
2. Enabling Secure AI-Assisted Knowledge Access in Federal Agencies: A Governance-Ready Retrieval-Augmented Generation (RAG) Architecture Validation, Sandeep Kumar Chittimalli, Publication date: 04/2026
Link: <https://smdi.com/why-rag-is-becoming-the-architecture-of-trusted-government-ai/>

Talks, Oral Presentations, Support, Posters and Invited

1. OpenCalValPlan: A Reproducible System for Multi-Sensor Calibration, Validation, and SNO Planning (Oral Presentation, Paper Accepted)
Location: IGARSS 2026, Washington, DC
Link: <https://2026.ieeeigarss.org/papers.php> **Aug 2026**
2. Dig South Tech Summit: Featured Speaker
Location: South Carolina, Charleston
Link: <https://digsouth.com/techsummit/#speakers> **June 2026**
3. TIC 2026: Technology & Innovation Conference (TIC) 2026: Keynote Speaker & Session Chair
Location: Virtual & Kuala Lumpur, Malaysia
Link: <https://ticresearch.com/>

4. Agentic RAG for Federal Decision Science: Applying LLMs to High-Stakes Program and Funding Decisions - Session Speaker
Location: IEEE Cloud Summit 2026 - Washington DC
Link: <https://www.ieeecloudsummit.org/> **June 2026**
5. 2nd International Conference on Recent Trends & Advances in Water Science Remote Sensing & Climate Change. (Abstract Acceptance, Presentation) **Feb 2026**
Location: IIT Jammu, India (Virtual)
Link: <https://sites.google.com/iitjammu.ac.in/wsrsec-2/home>
6. Teaching Machine Learning Techniques Using Agentic Coding, Earth Science Information Partners (ESIP) **Jan 2026**
Link: <https://zenodo.org/records/18405571>
7. GCSAYN Cohort 4 Internship and Mentorship 2025 Graduation Ceremony **Dec 2025**
Location: Virtual (Virtual Speaker)
Link: <https://csaynglobal.org/graduation-for-the-gcsayn-cohort-4-internship-and-mentorship-program/>
8. AGU 2025 (Speaker, Contributor) **Dec 2025**
Link: [Open Science Pavilion at AGU25 | AGU Open Science Leadership](#)
Session: <https://agu.confex.com/agu/agu25/meetingapp.cgi/Session/277762>
9. AGU 2025 Open Science & Data Expert Volunteer (Speaker, Contributor) **Dec 2025**
Link: [Data Help Desk - ESIP](#)
10. ISG ISRS National Symposium 2025 (Abstract Acceptance) **Nov 2025**
Location: India, IIIT Khargpur
Link: [ISG - ISRS National Symposium 2025](#)
11. Landsat/Sentinel-2 for aquatic science and applications: Where are we now and where we want to be? (Contributor) **Aug 2022**
Link: https://landsat.usgs.gov/landsat/landsat_science_team/2022-08-16_Pahlevan_AquaticApplications.pptx
12. Aquatic Science and Applications Enabled by the Landsat 8 Operational Land Imager **Feb 2018**
Location: Ocean Science, Portland, Oregon, USA.
Link: <https://agu.confex.com/agu/os18/meetingapp.cgi/Paper/303949>
13. Landsat-Sentinel-2 constellation for monitoring aquatic systems across the United States (Contributor) **Feb 2018**
Link: https://landsat.usgs.gov/landsat/landsat_science_team/2018-02-Pahlevan_SSAI.pdf
14. Geospatial Monitoring of the SDGs-Sharing experiences on indicator 6.6.1 on freshwater related ecosystems, and exploring opportunities for better monitoring of the land-water-ocean **Nov 2017**
Location: NASA GSFC, Greenbelt, Maryland USA.
Link: https://ggim.un.org/meetings/2017-Mexico/documents/NASA_EO4SDGs.pdf
15. Landsat-Sentinel-2 for Aquatic Science Data: Steps Towards Transitioning from Research to Operations **Nov 2017**
Location: Pecora 20-USGS, Sioux Falls, SD, USA.
Link : http://pecora.asprs.org/pecora20/wp-content/uploads/2017/11/Pecora-Final-Program-web_11-2-1.pdf
- Landsat-Sentinel constellation for regular monitoring of global water quality: current status & future needs (Contributor)
Location: USGS, Sioux Falls, SD, USA.
Link: https://landsat.usgs.gov/landsat/landsat_science_team/2017-07-Pahlevan_Aquatic.pdf
https://landsat.usgs.gov/landsat/landsat_science_team/LST-2017-07-files.zip (File Name : 2017-07_Day2_Pahlevan_Aquatic.pdf) **Jul 2017**
16. Reflectance-based Calibration of the Landsat Archive: Landsat-8 OLI to Landsat-1MSS (Contributor)
Location: CEOS WGCV IVOS Meeting Tucson, AZ, USA

- Link: [Reflectance-based Calibration of the Landsat Archive: Landsat-8 OLI to Landsat-1MSS](#),
[Alternate Link](#) **Mar 2017**
17. Reflectance-based Calibration of the Landsat Archive-Wrap-up
 Link: https://landsat.usgs.gov/landsat/landsat_science_team/2017-01-Helder_ReflectCalibration.pptx **Jan 2017**
 18. Reflectance-Based Cross-Calibration of Landsat Sensors Or... MSS Calibration: The Final Word!
Location: Landsat Science Team Meeting, McCrory Gardens Education and Visitor Center, Brookings, SD
 Link: [Reflectance-Based Cross-Calibration of Landsat Sensors Or... MSS Calibration: The Final Word!](#) **Jul 2016**
 19. USGS MSS Collection Plans and TIRS SSM Update **Jul 2016**
Location: USGS EROS, Sioux Falls, USA
 Link: https://landsat.usgs.gov/landsat/landsat_science_team/2016-07-Morfitt_MSS_Collection_Plans.pptx
 20. Maximizing the Value of the Landsat Archive for Forest Change Applications
Location: Landsat Science Team Meeting, McCrory Gardens Education and Visitor Center, Brookings, SD (Contributions) **Jul 2016**
 Link: https://landsat.usgs.gov/landsat/landsat_science_team/2016-07-Cohen-LandsatScienceTeamMeeting_%20July_2016_2.pptx
 21. Improvements in Landsat MSS data Radiometric Calibration **Jun 2016**
 (Oral Presentation)
Location: Landsat Technical Interchange Meeting-USGS, Sioux Falls, SD, USA.
 22. Reflectance based Cross-Calibration of MSS and TM sensors data-current status.
Location: Landsat Technical Interchange Meeting, Virginia Polytechnic Institute. USA.
 Link: https://landsat.usgs.gov/landsat/landsat_science_team/2016-01-MSS-Calibration-Comments.pptx, From Slide – 9 **Jan 2016**
 23. WiMAX Technology and its applications (Oral Presentation) **Aug 2011**
Location: Srinidhi Institute of Science and Technology, Hyderabad, India
 24. Nanotechnology and its impact on Integrated Chip Technology **Jul 2010**
 (Oral Presentation)
Location: JB institute of Engineering and Technology (JBIET), Hyderabad, India.

Peer Reviewer | Program Committee (PC) & International Program Committee (IPC) Member

National & International Conferences

- 13th IEEE Conference on Systems, Process & Control (ICSPC 2025) - Malacca, Malaysia
- 22nd IEEE International Colloquium on Signal Processing & Its Applications - Langkawi, Malaysia
- IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation Gwalior, India
- 7th International Symposium on Advanced Electrical and Communication Technologies (ISAECT 2026) – Mohali, India
- BDAA First International Conference on Big Data Analytics & Applications (BDAA) - Innsbruck, Austria
- IFIP TC11 Information Security & Privacy Conference (SEC) - Perth, Australia
- 2nd International Conference on Recent Trends & Advances in Water Science, Sensing & Climate Change - IIT Jammu, India

- 1st International Conference on Deep Learning Innovations for Smart Humanized AI - Jaipur & Delhi, India
- 4th International Joint Conference on Water Distribution Systems Analysis and Computing and Control in the Water Industry (WDSA/CCWI) – Cyprus
- SIWW2026 Water Convention - Singapore International Water Week (SIWW) - Singapore
- 17th IEEE Control & System Graduate Research Colloquium - USA
- IEEE International Conference on AI and Data Analytics (ICAD) - USA
- Computer Vision for Earth Observation Workshop Series – WACV - USA
- Icebound: 1st International Conference AI, EDGE & BLOCKCHAIN SYSTEMS - USA
- FLAIRS: The 39th International FLAIRS Conference - 2026 USA
- NASA Lifelines Earth Science Review Board, NASA Lifelines
- 2026 IEEE International Geoscience and Remote Sensing Symposium, DC, USA
- IEEE Technology & Innovation Conference (TIC 2026)
- Germany India Conference on Innovation, Technology, and Engineering (GICITE) 2026
- 2026 IEEE International Conference on Electronic Systems and Artificial Intelligence Applications (ICESAIA 2026)

Reviewer: Journals

- Springer Nature: Discover GeoScience
- Springer Nature: Discover Sustainability
- BHTY: Blockchain in Healthcare Today Journals
- FLAIRS: Florida Artificial Intelligence Research Society - Online Journals

Membership

- IEEE and IEEE GRSS Senior Member
- NASA Lifelines Earth Science Review Board, NASA Lifelines
- GCSAYN Governing Board member, Research & Scientific Core Advisory committee member
- AI Collective – Research Triangle Chapter Lead, North Carolina
- International Society for Photogrammetry and Remote Sensing (ISPRS)
- IET - Institution of Engineering and Technology - Hyderabad Local Chapter Advisor
- Institution of Engineering and Technology - IET
- Institute for Systems and Technologies of Information, Control and Communication (INSTICC)

Judging, Evaluation Role & Mentor roles

- Smart India Hackathon (SIH) – Judge, National Level Screening (India)
- SDG Design Challenge – Judge (USA)
- AI Collective – Judge & Chapter Lead (USA)
- Vibe Pitch – Judge, Live AI/Tech Startup Pitch Series
- Vibe Pitch 7 – Judge, Live AI/Tech Startup Pitch Series
- Harrisburg Vibes & n8n – Judge, Startup Pitch Competition (USA)
- Vibe Awards 2025 – Judge, AI/Web3 Startup Pitch Awards, USA
- Vibe AI Awards – Judge, AI Startup Awards, USA
- Hack-Nation Global AI Hackathon, MIT – Mentor (USA)
- Seattle Startup Summit Workshop – Judge, Startup Pitch Competition
- Reading-Berks Science & Engineering Fair – Judge, Science Fair (USA)
- NC Science and Engineering Fair – Judge (USA)
- Business Intelligence Group – Judge (USA)

Volunteer

- IEEE
- ESIP Open Science & Data help Desk
- AGU
- GCSAYN
- AI Collective
- IET

Technical Skills

- **Programming Languages:** Python, R, IDL, MATLAB, Shell Scripting, OpenCV, Pyspark.
- **Artificial Intelligence & Data Science:** Machine Learning, Deep Learning, Time Series Analysis, Predictive Analytics, Generative AI, Agentic AI
- **Statistical Packages:** Python (Pandas data Frame, Scikit, SciPy, NumPy), MATLAB, R-programming, MS Office Excel.
- **Web Services & Data Analytic Dashboard:** Tableau, OBIEE, and other data visualizations using python/R such as Streamlit, Voila, Bokeh, RShiny, Plotly Dash, Greppo & Datashader for geospatial web apps, HTML, Python Django.
- **Remote Sensing and GIS Tools:** ENVI, ArcGIS, QGIS, Rasterio, Shapely, Google Earth Engine (GEE), GeoPandas.
- **Databases:** Microsoft MySQL, Postgres SQL and PostGIS, Oracle and Oracle Spatial, SpatialLite with Sqlite, Redshift, Snowflake, Neo4j for graph and geospatial data, NoSql, Mongo DB and Data formats like JSON, XML, AVRO, Parquet, and spacecraft data such as Hierarchical Data Format (HDF4), Geotiff, JPEG, TIFF etc.).
- **Cloud knowledge:** Amazon Web services (AWS) Cloud.
- **Web Crawlers in Python:** Rest, Restful, SOAP.
- **Operating System:** UNIX, Linux, AWS EC2 cloud Platform, Windows, Mac and adaptable to any platform.
- **Computation Engine:** DataBricks, SLURM and High-Performance Computing (HPC) machines, AWS Services and cloud-based clusters.
- **Container:** Docker for deployments
- **IDEs for Analytics:** JupyterHub, Posit Workbench, PyCharm, Anaconda JupyterLab, Spyder, Eclipse, Atom, Notepad++, Pydev, PyStudio and Sublime text, ENVI.
- **Documentation:** Excellent documentation with Latex, R markdown and Jupyter Notebook.